Solving Difference Equations: A Worked Example

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Find $y[n]$ if $x[n] = 5\sin(\pi/2\, n)$, $y[-2] = 20$, $y[-1] = -2$ 1

$$y[n] - 1/4\, y[n-2] = x[n] - x[n-1]$$

1) Steady State Response

$(C_1 z_1^n + C_2 z_2^n)$

$z_1 = e^{j\pi/2}$, $z_2 = e^{-j\pi/2}$

$$x[n] = \frac{5}{2j} e^{j\frac{\pi}{2}n} - \frac{5}{2j} e^{-j\pi/2\, n}$$

$$y_s[n] = \frac{5}{2j}\left[H(e^{j\pi/2})\, e^{j\pi/2 n} - H(e^{-j\pi/2})\, e^{-j\pi/2\, n} \right]$$

$$H(z) = \frac{1 - z^{-1}}{1 - \frac{1}{4}z^{-2}} \quad \text{so}$$

$$H(e^{j\pi/2}) = \frac{1 - e^{-j\pi/2}}{1 - 1/4\, e^{-j\pi}} = \frac{1 + j}{5/4} = \frac{4}{5}(1+j)$$

$$H(e^{-j\pi/2}) = \frac{1 - e^{j\pi/2}}{1 - 1/4\, e^{j\pi}} = \frac{4}{5}(1-j)$$

So $$y_s[n] = \frac{5}{2j}\left[\frac{4}{5}(1+j)\,e^{+j\pi/2\,n} - \frac{4}{5}(1-j)\,e^{-j\pi/2\,n}\right]$$

$$= 4\left\{\frac{e^{j\pi/2 n} - e^{-j\pi/2\,n}}{2j} + \frac{e^{j\pi/2 n} + e^{-j\pi/2\,n}}{2}\right\}$$

$$= 4\sin\left(\frac{\pi}{2}n\right) + 4\cos(\pi/2\,n)$$

2) Transient Response

$$y_t[n] = \sum_{\ell=1}^{2} c_\ell\, d_\ell^{\,n}, \quad d_\ell \text{ roots of } \quad 1 - 1/4\, z^{-2} = 0$$

or $z^2 - 1/4 = 0 \Rightarrow d_1 = 1/2, d_2 = -1/2$

$$y_t[n] = c_1\,(1/2)^n + c_2\,(-1/2)^n$$

3) Complete Solution

$$y[n] = y_s[n] + y_t[n] = 4\sin\left(\tfrac{\pi}{2}n\right) + 4\cos\left(\tfrac{\pi}{2}n\right) + c_1\left(\tfrac{1}{2}\right)^n + c_2\left(-\tfrac{1}{2}\right)^n \quad (n \geq 0)$$

– Use initial conditions to find c_1, c_2

translate initial conditions forward

$$y[0] = \tfrac{1}{4}y[-2] + x[0] - x[-1] = \tfrac{1}{4}20 + 0 = 5$$

$$y[1] = \tfrac{1}{4}y[-1] + x[1] - x[0] = \tfrac{1}{4}(-2) + 5 - 0 = 4.5$$

Thus,

$$5 = 4\sin\left(\tfrac{\pi}{2}\cdot 0\right) + 4\cos\left(\tfrac{\pi}{2}\cdot 0\right) + c_1 + c_2 \Rightarrow 1 = c_1 + c_2$$

$$4.5 = 4\sin\left(\tfrac{\pi}{2}\cdot 1\right) + 4\cos\left(\tfrac{\pi}{2}\cdot 1\right) + \tfrac{1}{2}c_1 - \tfrac{1}{2}c_2 \Rightarrow \tfrac{1}{2} = \tfrac{1}{2}c_1 - \tfrac{1}{2}c_2$$

$\Rightarrow c_1 = 1, \; c_2 = 0$

$$y[n] = 4\sin\left(\tfrac{\pi}{2}n\right) + 4\cos\left(\tfrac{\pi}{2}n\right) + \left(\tfrac{1}{2}\right)^n$$